

AMITH LAL

AI & Machine Learning Engineer | Generative AI, RAG Systems & LLM Applications | Python · Django

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PROFESSIONAL SUMMARY

Generative AI and Machine Learning Engineer with strong Python skills, focused on clean, secure, and maintainable code. Hands-on in designing and testing **LLMs and prompts** and implementing **RAG-based systems** end to end — chunking, embedding generation, and vector search. Experienced in creating and maintaining API endpoints and system architecture with a full **Gen AI stack (local LLMs, vector DBs, LangChain, Ollama, n8n)**. AWS certified, with solid SQL and ML knowledge plus Git, CI/CD, code reviews, Docker, and Kubernetes.

CORE COMPETENCIES

Gen AI Stack	Local LLMs (Qwen, Gemma, Llama) · Ollama · Vector DBs (FAISS, ChromaDB) · LangChain · n8n · RAG · Chunking · Embeddings · Vector Search · Prompt Design & Testing · Agentic Workflows · Claude Code · MCP Servers
Machine Learning	Scikit-learn · Predictive Modeling · Model Deployment · Data Pipelines
Languages	Python (strong) · SQL · Java
Backend & APIs	Django · Django REST Framework · FastAPI · RESTful APIs · API Design & Maintenance · System Architecture · Firebase
DevOps & Cloud	AWS (EC2, S3, IAM) – AWS Certified · Vercel · Docker · Kubernetes · CI/CD · Git / GitHub · Code Reviews
SAP Technologies	ABAP · CDS Views · OData · FPM · Fiori · UI5
Databases & Storage	Supabase (PostgreSQL) · MySQL · SQLite · Firestore
Engineering Practices	Clean, Secure & Maintainable Code · OOP · Data Structures & Algorithms · API Integration · Postman · Jupyter

PROFESSIONAL EXPERIENCE

AI and Machine Learning Engineer

Jul 2024 – Present

Linx AS, Kochi, India

Gen AI Document Processing & Extraction Platform | Python, Django, Ollama, LLMs, RAG

- Developed a Django-based **Generative AI** application leveraging **local LLMs (Qwen, Gemma, Llama)** via **Ollama** and **RAG** to ingest, analyze, and extract structured data from diverse formats (PDF, Word, Excel, CSV, JSON).
- Implemented **RAG-based pipelines** end to end — document chunking, embedding generation, and vector search (FAISS/ChromaDB) — enabling accurate Q&A, AI summaries, and targeted data extraction from uploaded files.
- Fine-tuned LLM models using Supervised Fine-Tuning (SFT) and LoRA (Low-Rank Adaptation) techniques to improve domain-specific accuracy and reduce inference/compute costs.
- Designed and iteratively **tested LLM prompts** to improve extraction accuracy and response reliability.
- Created and maintained robust **RESTful API endpoints** (Django REST Framework) for file uploads and extraction, owning system architecture and integrating directly with SAP systems.
- Used **Claude Code** for AI-assisted development; wrote clean, secure, and maintainable code with Git best practices and **code reviews**.

Albert Invent Data Migration & Utility Development | Python, Albert SDK, Pandas

- Executed complex data migration initiatives transferring critical R&D data from diverse client legacy systems into the Albert Invent platform.
- Developed custom utility tools with Python and the Albert SDK to automate data extraction, validation, and loading; engineered efficient bulk-processing scripts ensuring high data integrity and minimal downtime.

Predictive Color Mixing Platform | Python, Machine Learning, Scikit-learn

- Engineered and deployed a predictive ML model using **Scikit-learn** to optimize industrial color mixing, significantly improving formulation accuracy and efficiency.

Cloud Engineering Intern

Feb 2024 – May 2024

Trainso Training Solutions, Bengaluru, India

- Worked on AWS cloud infrastructure operations (IAM, EC2, S3, security configurations); assisted in deploying containerized (Docker) applications and managing access control policies.

EDUCATION

B.Tech – Computer Science · APJ Abdul Kalam Technological University · CGPA: 9.34

2019 – 2023

CERTIFICATIONS & PUBLICATIONS

ML Specialization, Andrew Ng (2024) · IBM Applied AI Professional Certificate (2023) · Python for Data Science & AI, IBM (2023) · AWS Solutions Architect Associate (2023) · Publication: Personalizable Super-App Framework, NCASCD (2023)